

GSVA for mutil Group

Ximing Ran

2025-03-24

Contents

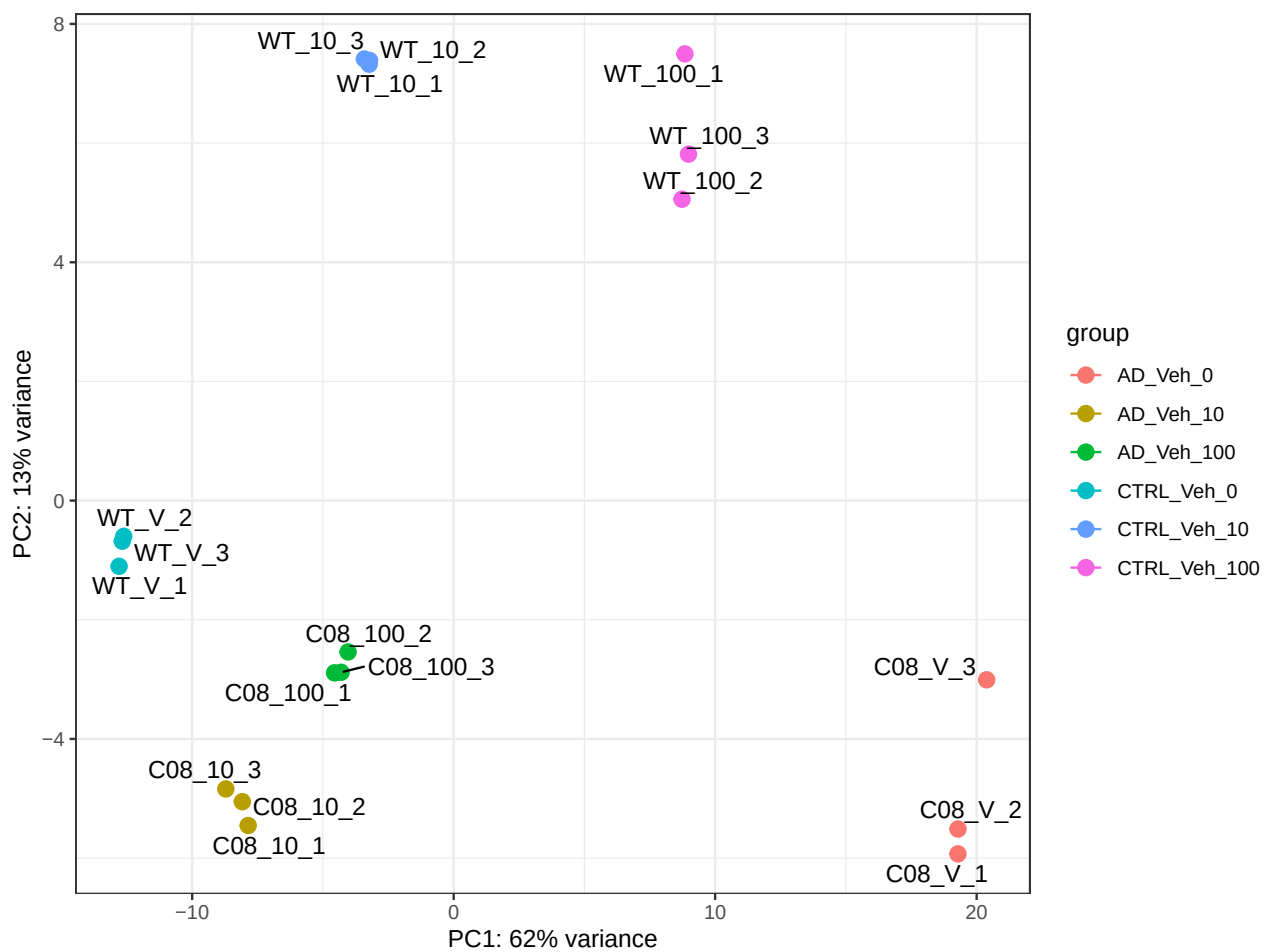
1. Read the count data	2
2. Visualization for reuslt	3
(1) Sample information - PCA plot	3
(2) Sample information - Distance heatmap	4
3. GSVA analysis	5
Select the significant pathway	10
Session information	15

1. Read the count data

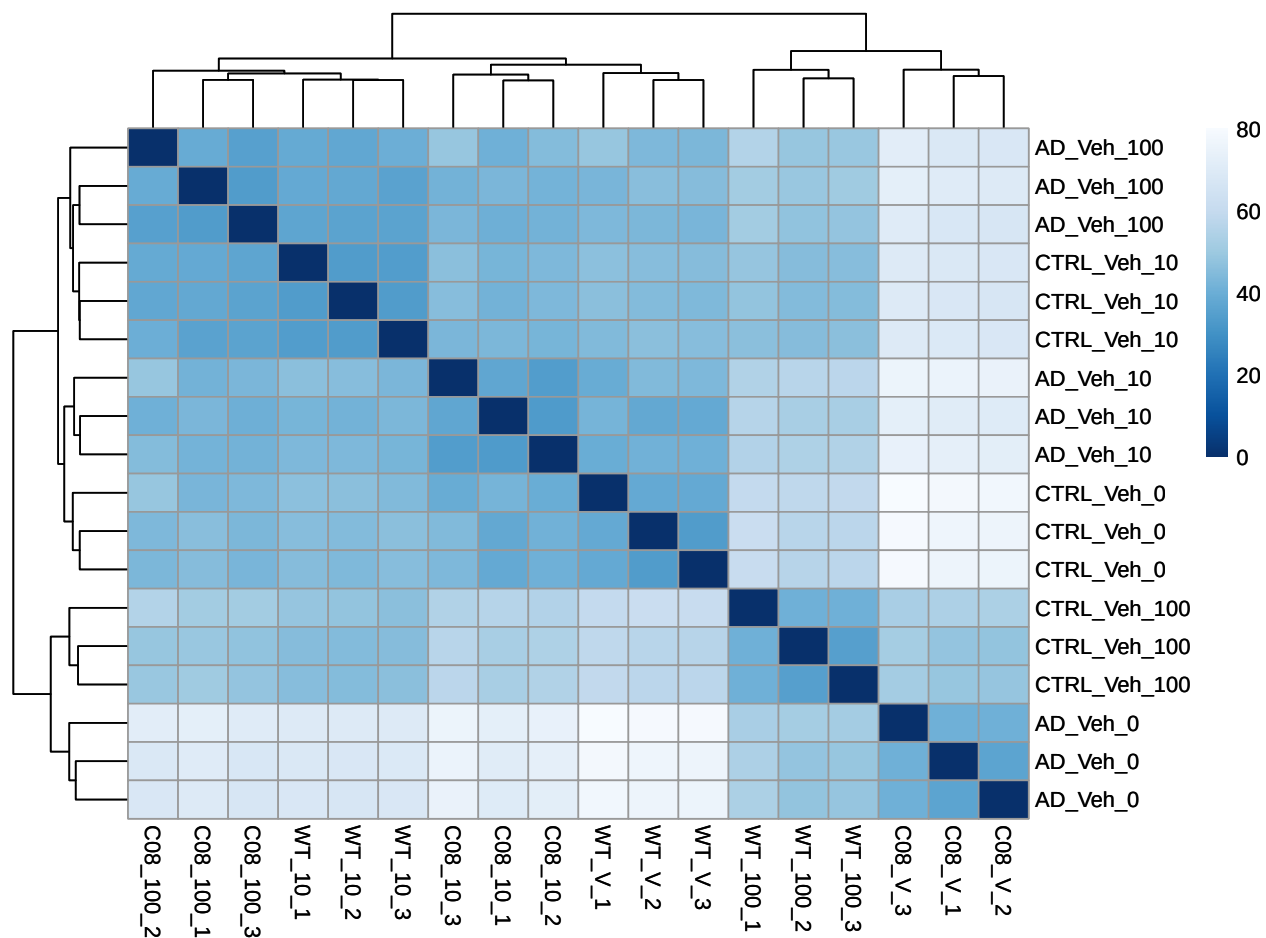
In this section, we will read the clean count data from the `synaptosomes_bulkRNA` folder. We will read the data and merge them into a single table.

2. Visualization for reuslt

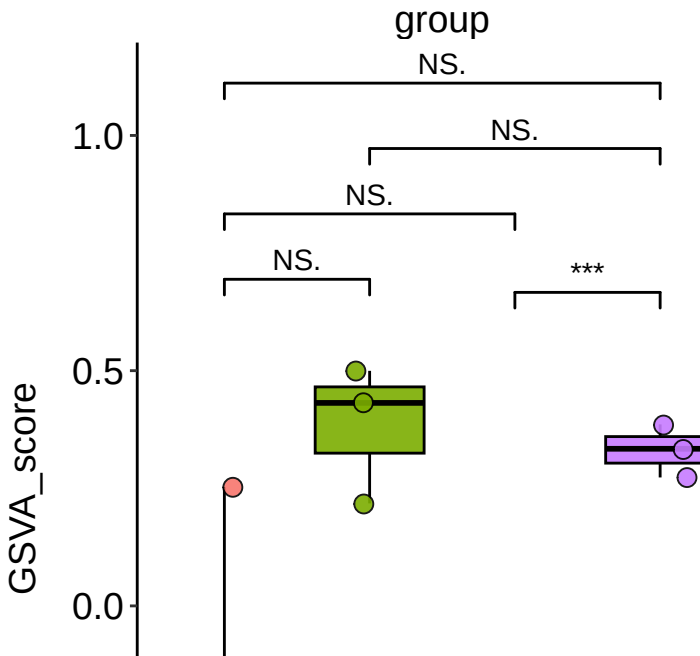
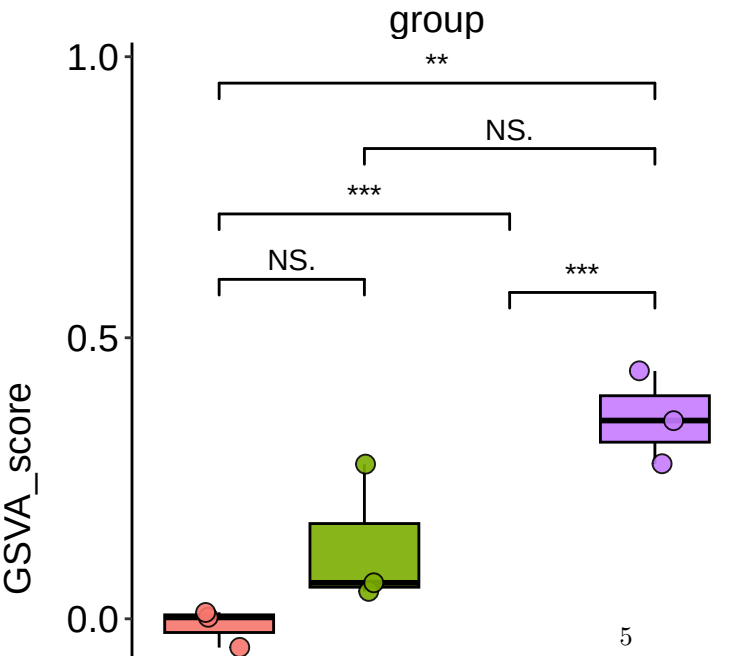
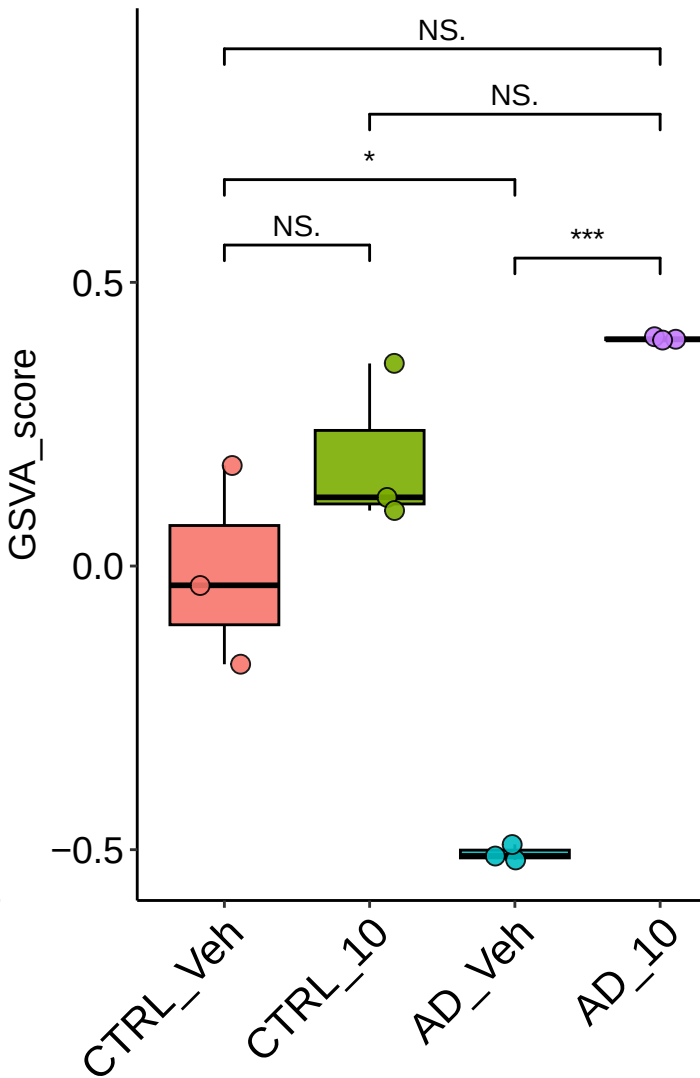
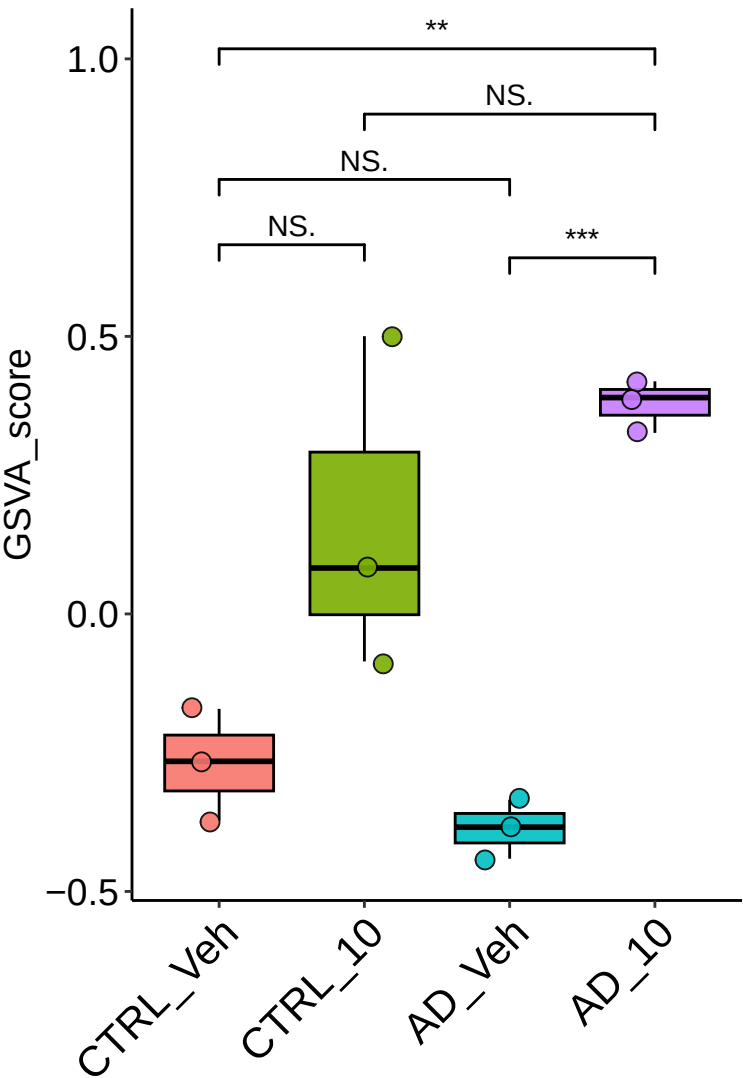
(1) Sample information - PCA plot

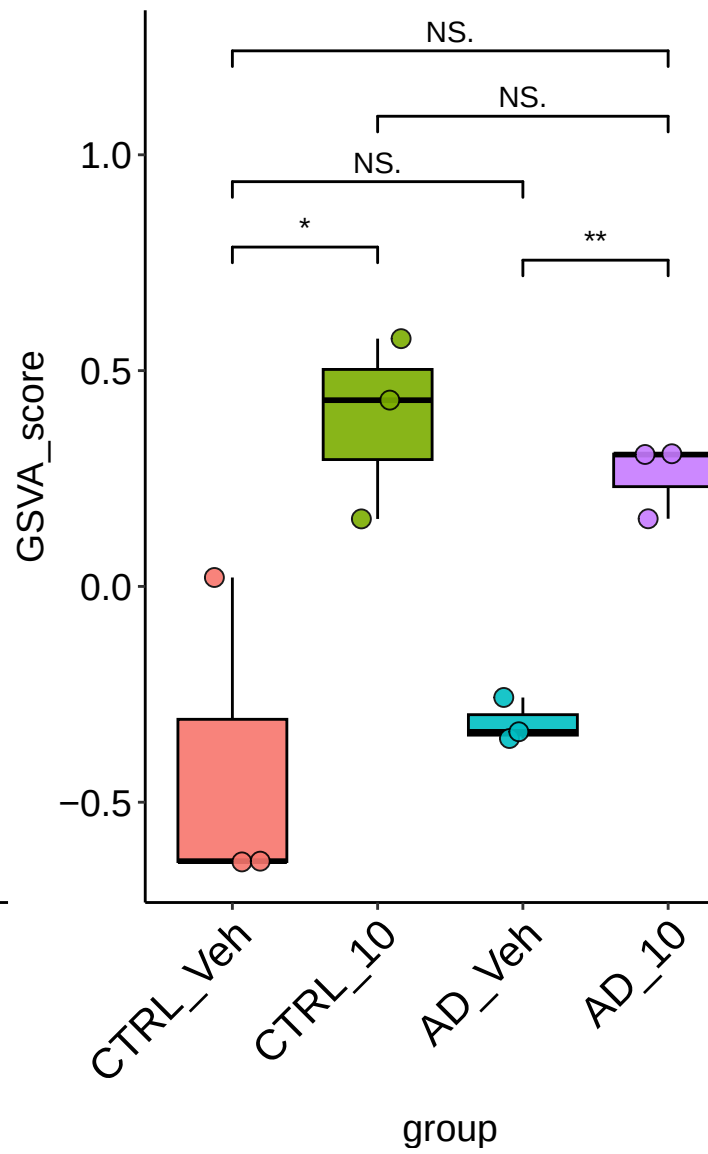
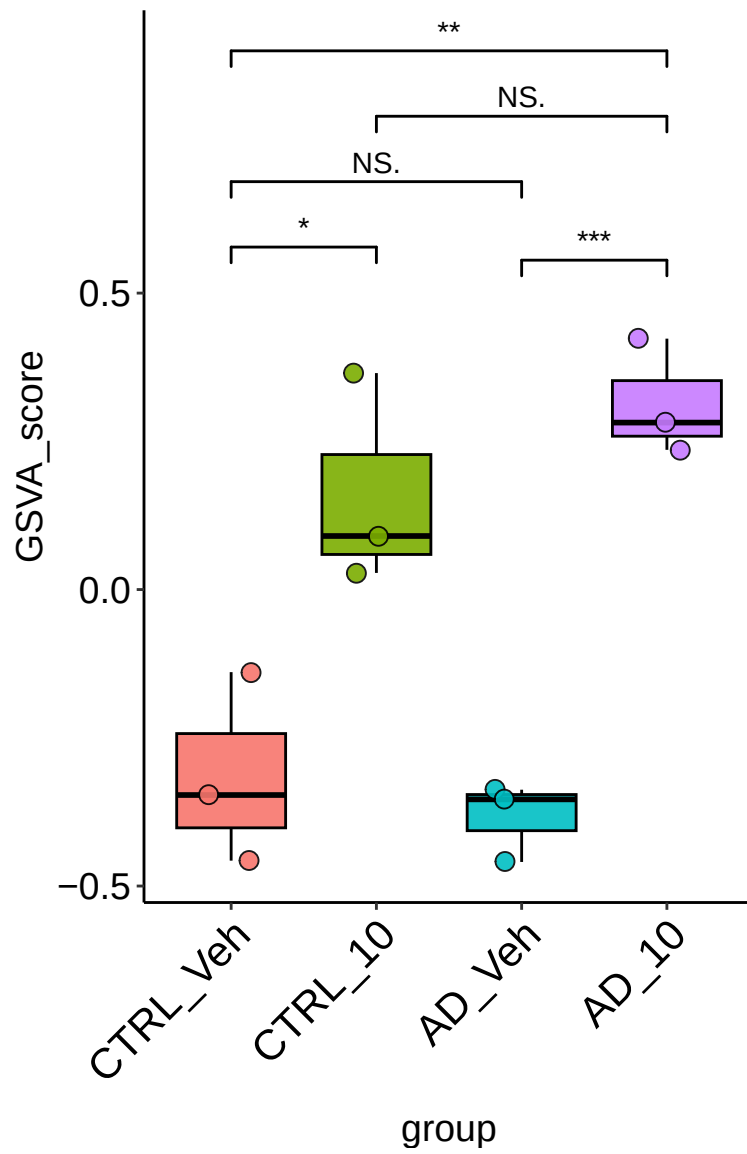


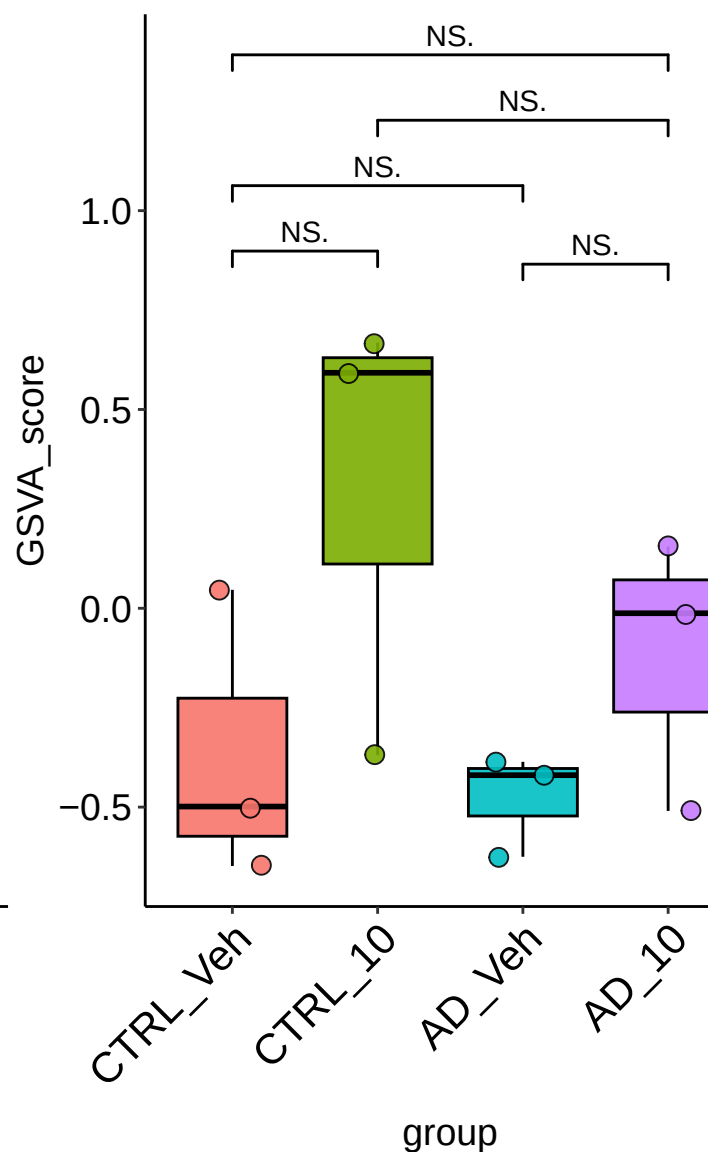
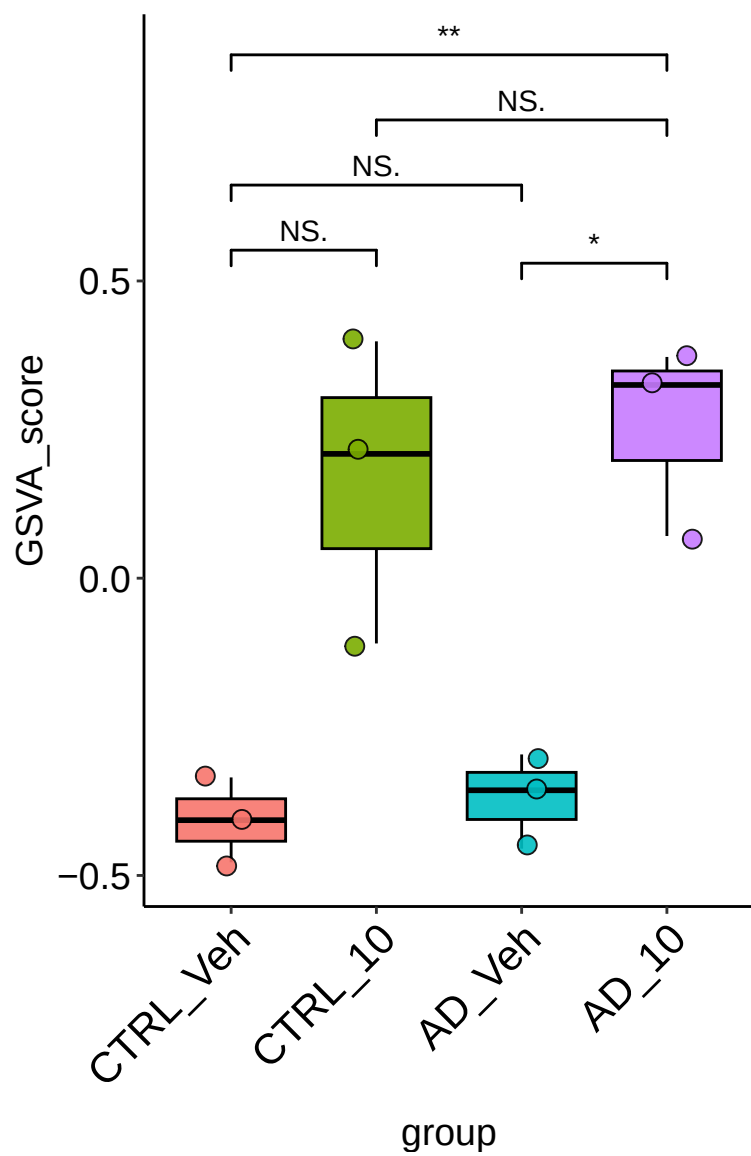
(2) Sample information - Distance heatmap

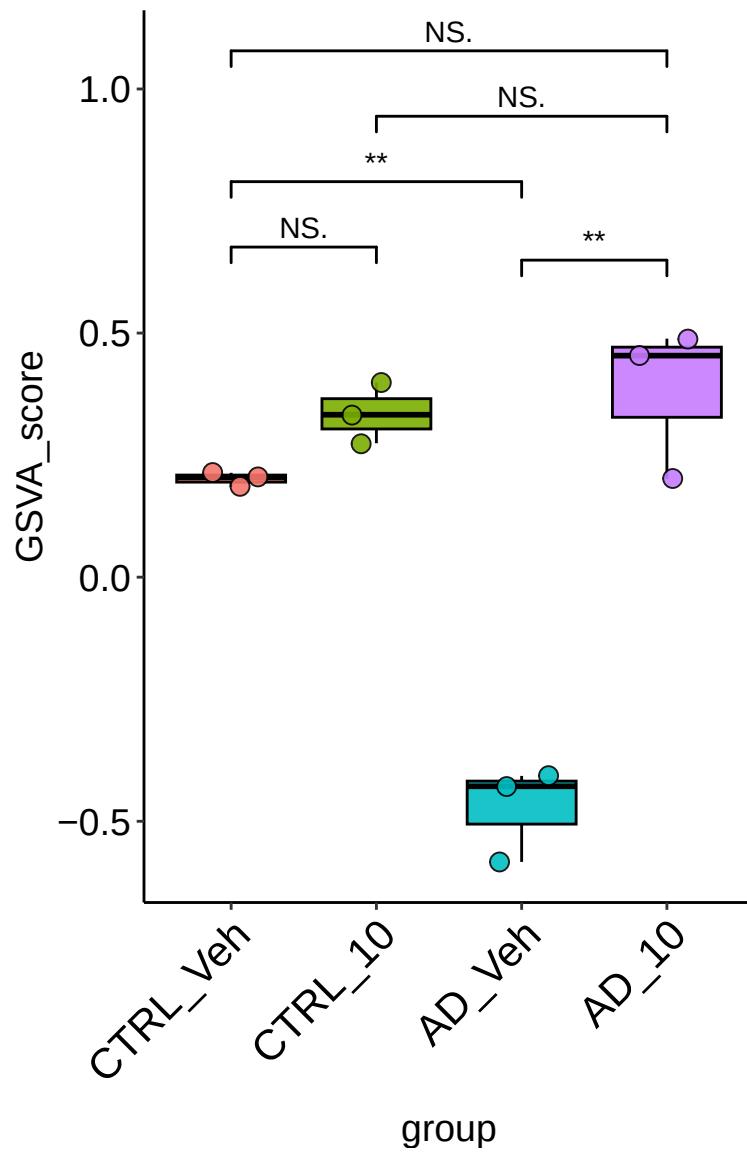


3. GSVA analysis

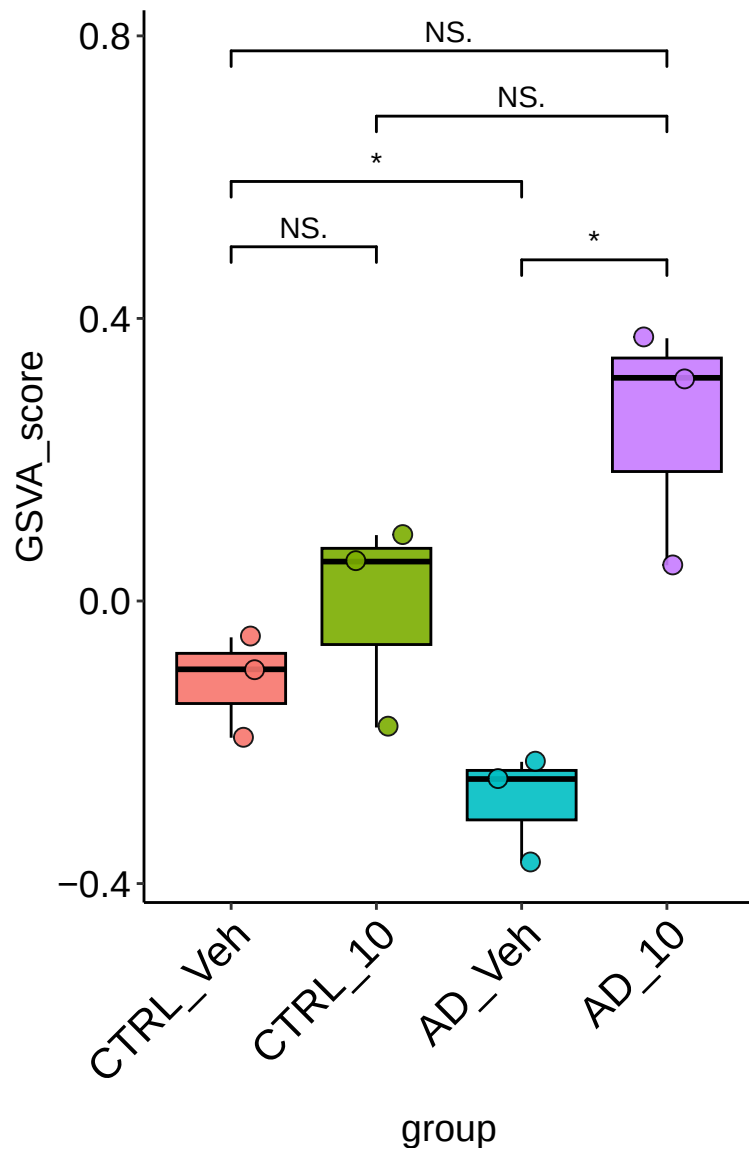




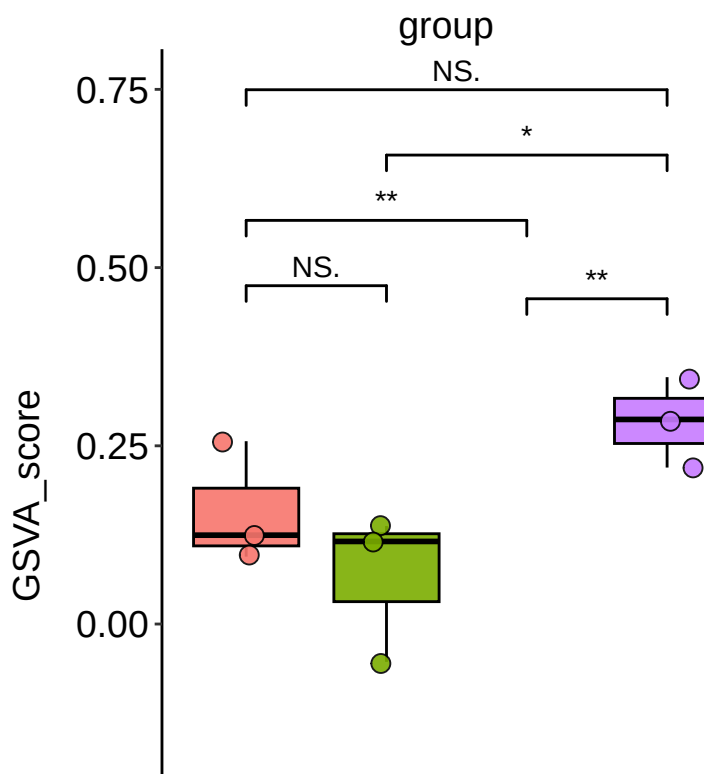
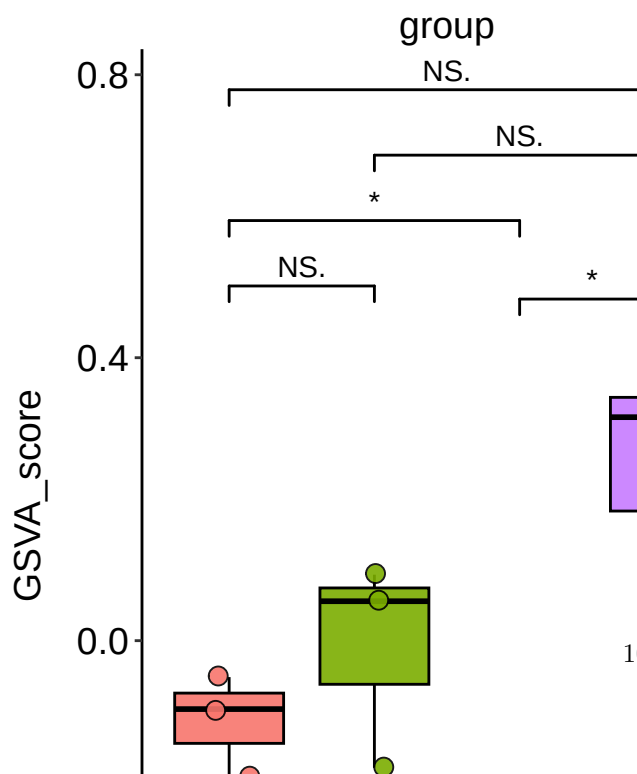
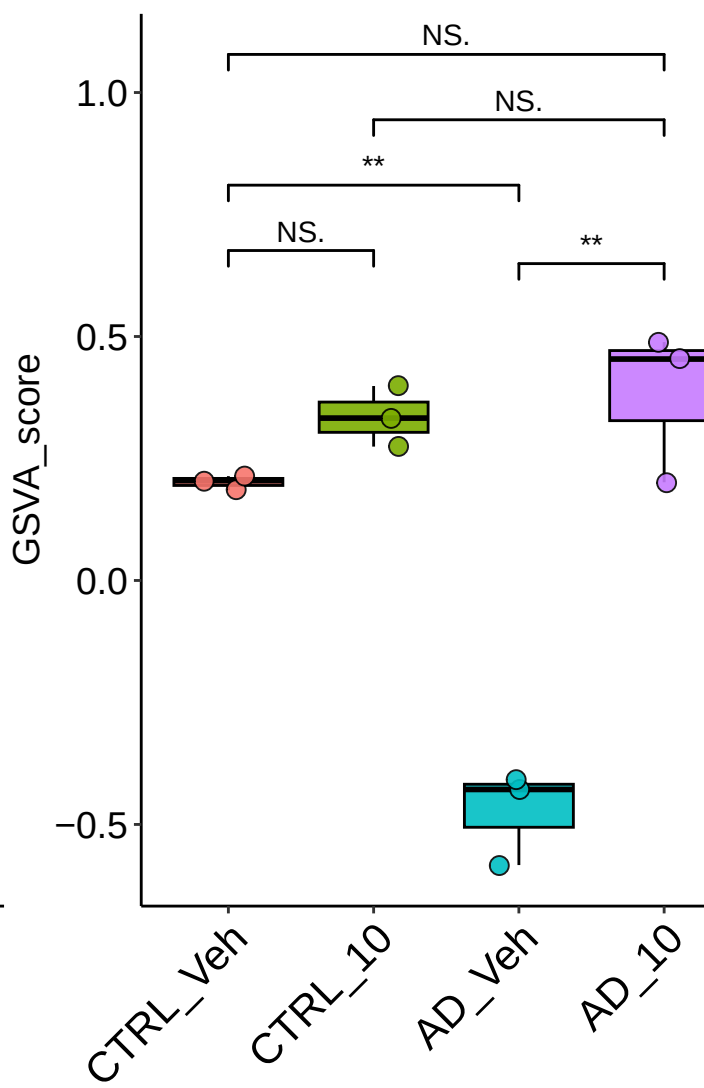
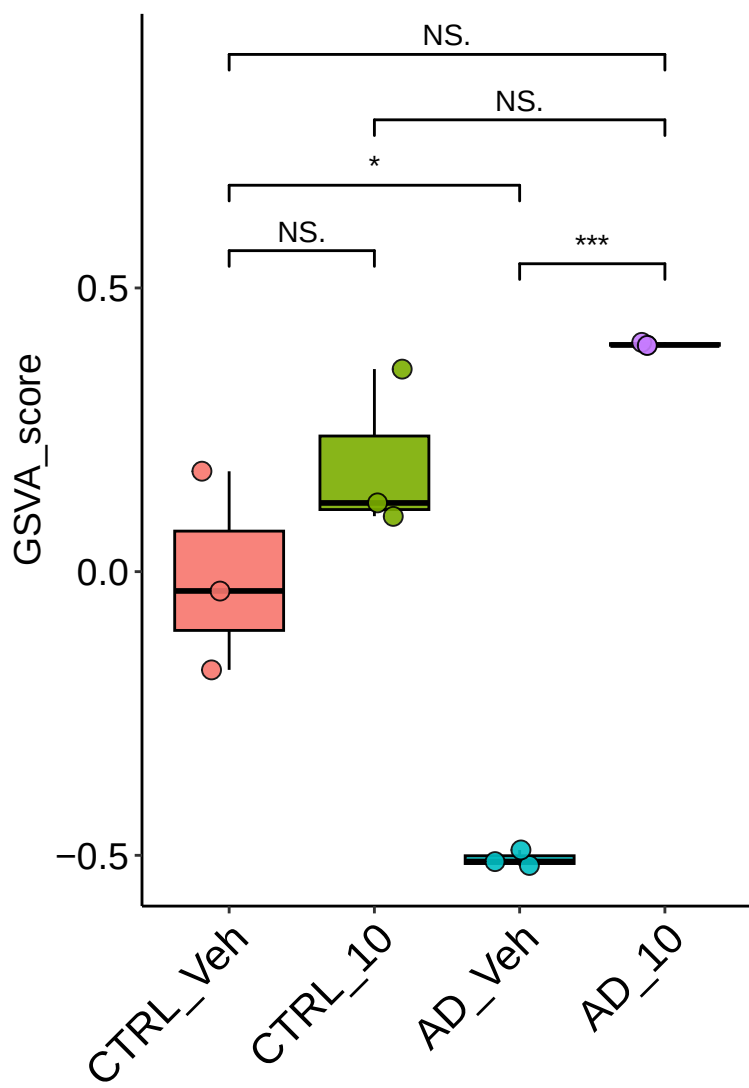


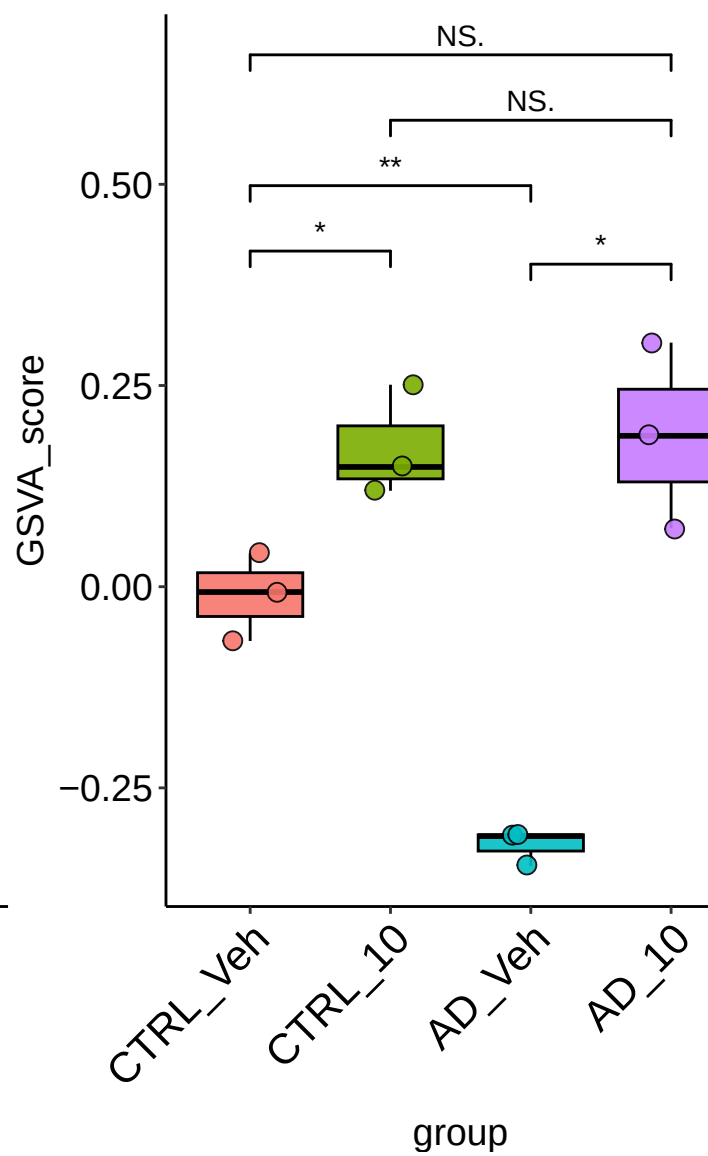
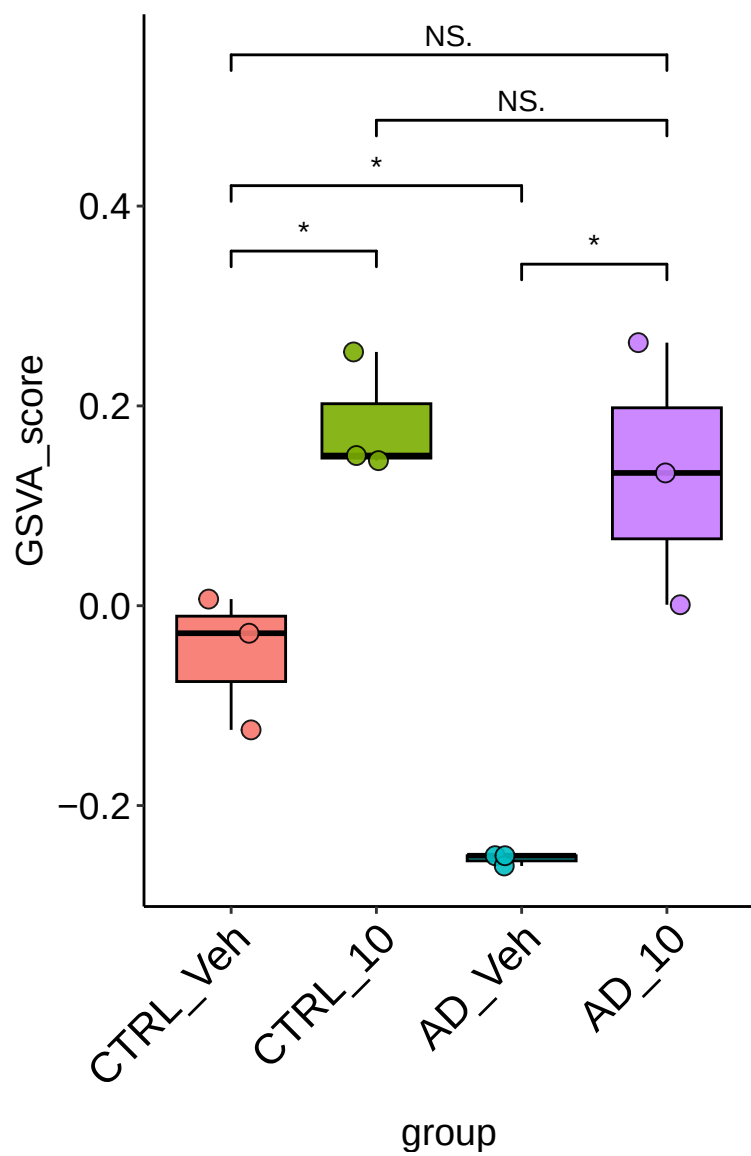


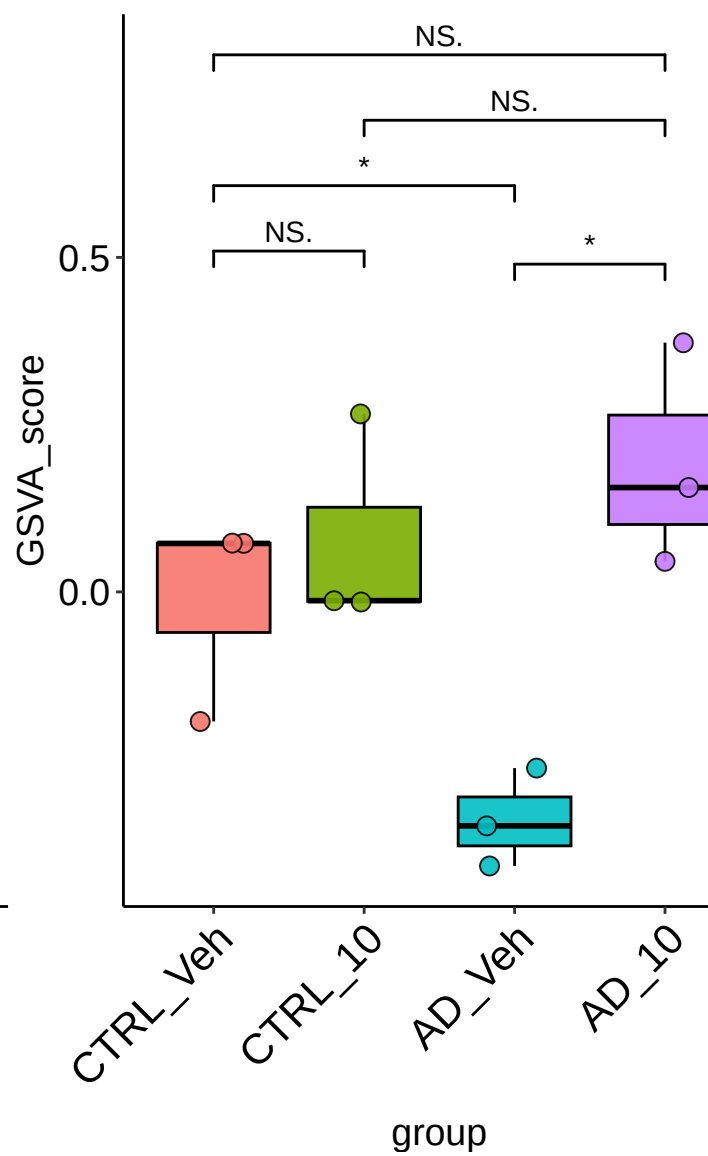
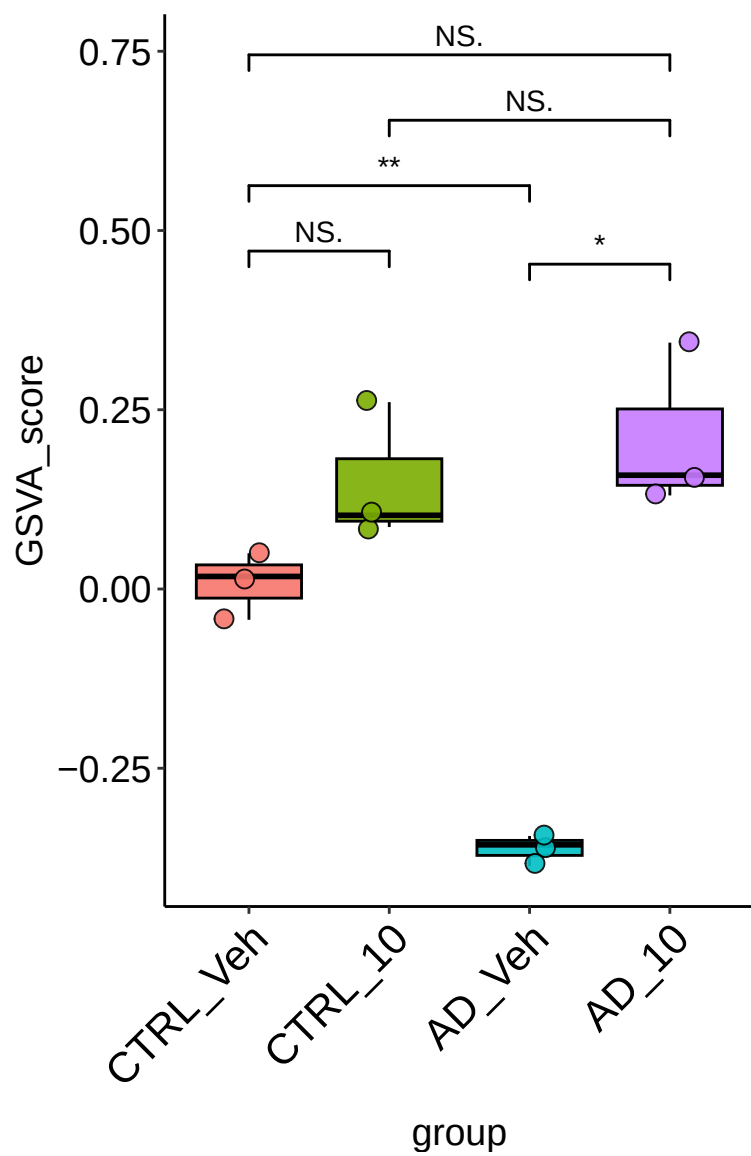
[1] "Processing 10 pathway"

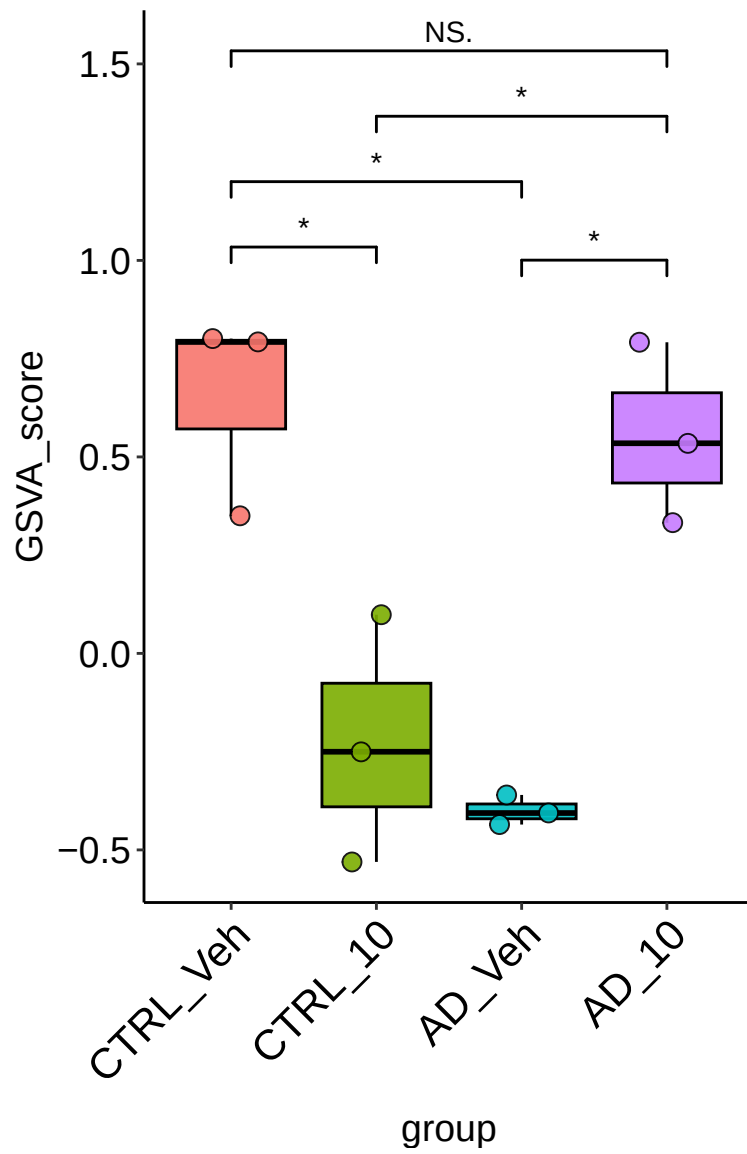


Select the significant pathway

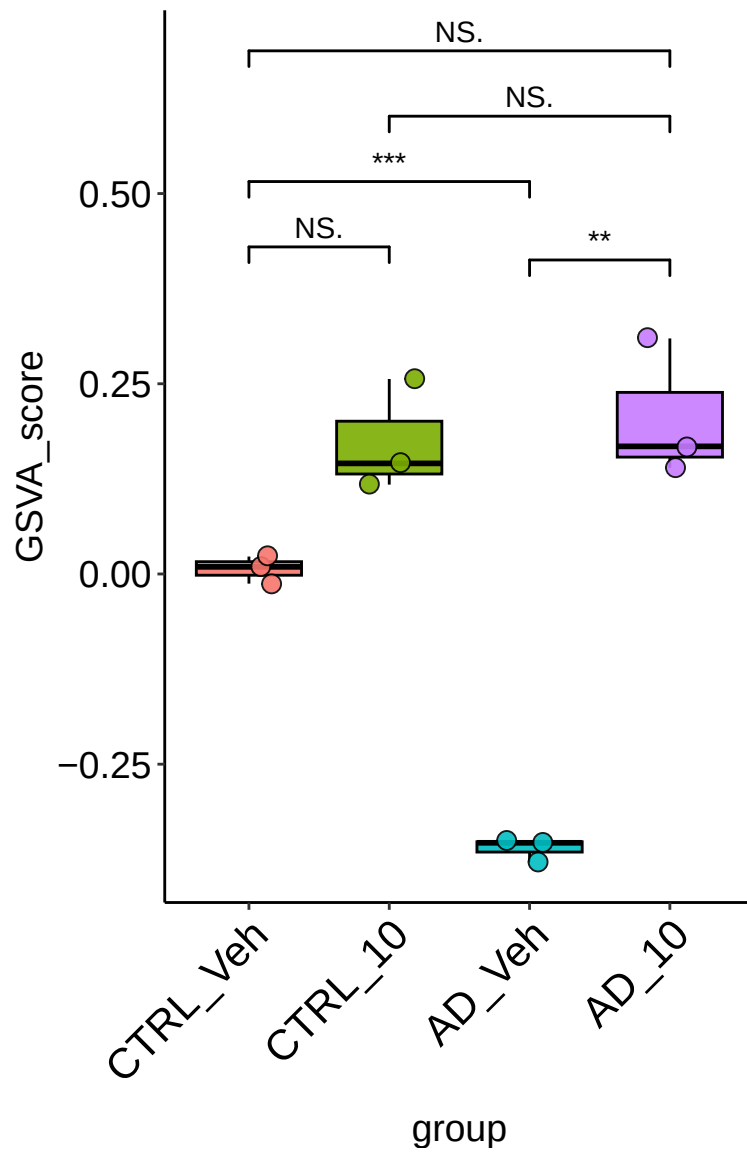








[1] "Processing 10 pathway"



Session information

```
## R version 4.4.0 (2024-04-24)
## Platform: aarch64-apple-darwin20
## Running under: macOS Sonoma 14.3.1
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib; LAPACK v
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] parallel stats4 stats graphics grDevices utils datasets
## [8] methods base
##
## other attached packages:
## [1] GSEABase_1.66.0 graph_1.82.0
## [3] annotate_1.82.0 XML_3.99-0.18
## [5] extrafont_0.19 ggsignif_0.6.4
## [7] patchwork_1.3.0 decoupleR_2.10.0
## [9] GSVA_1.52.3 BiocParallel_1.38.0
## [11] edgeR_4.2.2 limma_3.60.6
## [13] GenomicFeatures_1.56.0 biomaRt_2.60.1
## [15] gprofiler2_0.2.3 RColorBrewer_1.1-3
## [17] data.table_1.16.4 org.Hs.eg.db_3.19.1
## [19] AnnotationDbi_1.66.0 clusterProfiler_4.12.6
## [21] ggfortify_0.4.17 pheatmap_1.0.12
## [23] EnhancedVolcano_1.22.0 ggrepel_0.9.6
## [25] apeglm_1.26.1 DESeq2_1.44.0
## [27] SummarizedExperiment_1.34.0 Biobase_2.64.0
## [29] MatrixGenerics_1.16.0 matrixStats_1.5.0
## [31] reshape2_1.4.4 Matrix_1.7-2
## [33] Signac_1.14.0 Seurat_5.2.1
## [35] SeuratObject_5.0.2 sp_2.2-0
## [37] rtracklayer_1.64.0 GenomicRanges_1.56.2
## [39] GenomeInfoDb_1.40.1 IRanges_2.38.1
## [41] S4Vectors_0.42.1 BiocGenerics_0.50.0
## [43] knitr_1.49 lubridate_1.9.4
## [45] forcats_1.0.0 stringr_1.5.1
## [47] dplyr_1.1.4 purrr_1.0.4
## [49] readr_2.1.5 tidyr_1.3.1
## [51] tibble_3.2.1 ggplot2_3.5.1
## [53] tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] SpatialExperiment_1.14.0 R.methodsS3_1.8.2
## [3] progress_1.2.3 goftest_1.2-3
## [5] HDF5Array_1.32.1 Biostrings_2.72.1
## [7] vctrs_0.6.5 spatstat.random_3.3-2
```

## [9] digest_0.6.37	png_0.1-8
## [11] deldir_2.0-4	parallelly_1.42.0
## [13] magick_2.8.5	MASS_7.3-64
## [15] httpuv_1.6.15	qvalue_2.36.0
## [17] withr_3.0.2	xfun_0.51
## [19] ggfun_0.1.8	survival_3.8-3
## [21] memoise_2.0.1	gson_0.1.0
## [23] tidytree_0.4.6	zoo_1.8-12
## [25] pbapply_1.7-2	R.oo_1.27.0
## [27] prettyunits_1.2.0	KEGGREST_1.44.1
## [29] promises_1.3.2	httr_1.4.7
## [31] restfulr_0.0.15	rhdf5filters_1.16.0
## [33] globals_0.16.3	fitdistrplus_1.2-2
## [35] rhdf5_2.48.0	rstudioapi_0.17.1
## [37] UCSC.utils_1.0.0	miniUI_0.1.1.1
## [39] generics_0.1.3	DOSE_3.30.5
## [41] curl_6.2.1	zlibbioc_1.50.0
## [43] ScaledMatrix_1.12.0	ggraph_2.2.1
## [45] polyclip_1.10-7	GenomeInfoDbData_1.2.12
## [47] SparseArray_1.4.8	xtable_1.8-4
## [49] evaluate_1.0.3	S4Arrays_1.4.1
## [51] BiocFileCache_2.12.0	hms_1.1.3
## [53] irlba_2.3.5.1	colorspace_2.1-1
## [55] filelock_1.0.3	ROCR_1.0-11
## [57] reticulate_1.40.0	spatstat.data_3.1-4
## [59] magrittr_2.0.3	lmtest_0.9-40
## [61] later_1.4.1	viridis_0.6.5
## [63] ggtree_3.12.0	lattice_0.22-6
## [65] spatstat.geom_3.3-5	future.apply_1.11.3
## [67] scattermore_1.2	shadowtext_0.1.4
## [69] cowplot_1.1.3	RcppAnnoy_0.0.22
## [71] pillar_1.10.1	nlme_3.1-167
## [73] compiler_4.4.0	beachmat_2.20.0
## [75] RSpectra_0.16-2	stringi_1.8.4
## [77] tensor_1.5	GenomicAlignments_1.40.0
## [79] plyr_1.8.9	crayon_1.5.3
## [81] abind_1.4-8	BiocIO_1.14.0
## [83] gridGraphics_0.5-1	emdbbook_1.3.13
## [85] locfit_1.5-9.11	graphlayouts_1.2.2
## [87] bit_4.5.0.1	fastmatch_1.1-6
## [89] codetools_0.2-20	BiocSingular_1.20.0
## [91] plotly_4.10.4	mime_0.12
## [93] splines_4.4.0	Rcpp_1.0.14
## [95] fastDummies_1.7.5	sparseMatrixStats_1.16.0
## [97] dbplyr_2.5.0	Rttf2pt1_1.3.12
## [99] blob_1.2.4	here_1.0.1
## [101] fs_1.6.5	listenv_0.9.1
## [103] ggplotify_0.1.2	statmod_1.5.0
## [105] tzdb_0.4.0	tweenr_2.0.3
## [107] pkgconfig_2.0.3	tools_4.4.0
## [109] cachem_1.1.0	RSQLite_2.3.9
## [111] viridisLite_0.4.2	DBI_1.2.3
## [113] numDeriv_2016.8-1.1	fastmap_1.2.0
## [115] rmarkdown_2.29	scales_1.3.0

## [117] grid_4.4.0	ica_1.0-3
## [119] Rsamtools_2.20.0	coda_0.19-4.1
## [121] dotCall64_1.2	RANN_2.6.2
## [123] farver_2.1.2	tidygraph_1.3.1
## [125] scatterpie_0.2.4	yaml_2.3.10
## [127] cli_3.6.4	lifecycle_1.0.4
## [129] uwot_0.2.2	mvtnorm_1.3-3
## [131] timechange_0.3.0	gtable_0.3.6
## [133] rjson_0.2.23	ggridges_0.5.6
## [135] progressr_0.15.1	ape_5.8-1
## [137] jsonlite_1.9.0	RcppHNSW_0.6.0
## [139] bitops_1.0-9	bit64_4.6.0-1
## [141] Rtsne_0.17	yulab.utils_0.2.0
## [143] spatstat.utils_3.1-2	bdsmatrix_1.3-7
## [145] GOSemSim_2.30.2	spatstat.univar_3.1-1
## [147] R.utils_2.12.3	lazyeval_0.2.2
## [149] shiny_1.10.0	htmltools_0.5.8.1
## [151] enrichplot_1.24.4	GO.db_3.19.1
## [153] sctransform_0.4.1	rappdirs_0.3.3
## [155] tinytex_0.55	glue_1.8.0
## [157] spam_2.11-1	httr2_1.1.0
## [159] XVector_0.44.0	RCurl_1.98-1.16
## [161] rprojroot_2.0.4	treeio_1.28.0
## [163] gridExtra_2.3	extrafontdb_1.0
## [165] igraph_2.1.4	R6_2.6.1
## [167] SingleCellExperiment_1.26.0	labeling_0.4.3
## [169] RcppRoll_0.3.1	cluster_2.1.8
## [171] bbmle_1.0.25.1	Rhdf5lib_1.26.0
## [173] aplot_0.2.4	DelayedArray_0.30.1
## [175] tidyselect_1.2.1	ggforce_0.4.2
## [177] xml2_1.3.6	future_1.34.0
## [179] rsvd_1.0.5	munsell_0.5.1
## [181] KernSmooth_2.23-26	htmlwidgets_1.6.4
## [183] fgsea_1.30.0	rlang_1.1.5
## [185] spatstat.sparse_3.1-0	spatstat.explore_3.3-4